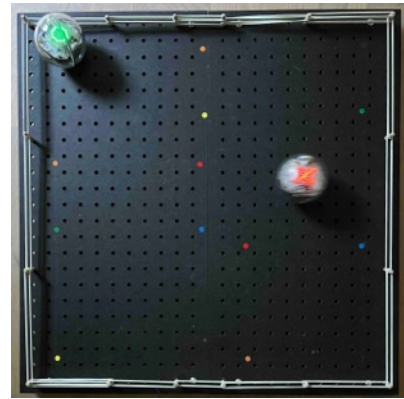


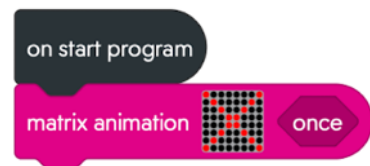


# Run Robot Run

*Robots give chase but may have to escape.*



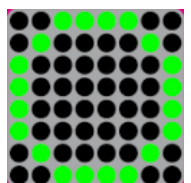
- 1) In Chrome, open <https://edu.sphero.com/code>
- 2) Connect your **Sphero BOLT** robot: click 
- 3) Select  then 
- 4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then 
- 5) Check your robot is showing the right image then 
- 6) Click **+ CREATE PROGRAM.** 
- 7) Input a name for your program. Leave **BLOCKS** selected.  
Select **SPHERO BOLT**, **deselect SPHERO BOLT+** then **CREATE.**
- 8) Add code for a **matrix animation** with *your own* design for a robot chaser in **red**.  
My example is a red **X**.



*Click on the matrix grid to edit then  to add a new 8x8 matrix, or  to edit an existing one. Click  to finish.*

 *the code. Does it work?*

- 9) Click on the **matrix** grid and press  to add another 8x8 matrix in **green** for when your robot has to escape. My example is a green **O**.



10) In **Variables**, create a new variable called **chase**.

This is **boolean** because it will only be *true* or *false*.

*Sphero robots communicate using infra-red light invisible to us. Chasers broadcast their position on channels 0 & 1, and those trying to escape on channels 2 & 3.*

11) Add code (right) that **follows** or **evades** these channels depending on **if** it has to **chase** or escape. Click on *true* (chase) or *false* (escape) to toggle it.

12) Sometimes the Sphero gets stuck on the sidelines.

You can add

code (left) after

the delay, inside the loop, to randomly spin Sphero if it stops.

13) We can toggle between chase and escape modes **on collision** (but it won't detect glancing blows). The code (right) toggles between true and false.

*To **Save** your code go back to the main page. Click on the 3 dots (ellipsis) next to your program and download to a USB drive.*

